
Sensitivity analyses to account for impact of rescue therapy

1 Appendix

1.1 Introduction

In the proposed phase 3 study, subjects' daily diary reported dysphagia day (DD) outcomes will be normalized based on available records over 14-day periods. Ideally there would be complete records for all subjects, however, in practice there will be missing daily records. Based on simulations, which are also utilized and described here, we found a minimum of 8 records per subject to be required to keep the loss in power for the primary analysis to be less than 5% relative to the full data (complete 14-day records) scenario. In order to minimize missing data due to DD collection the protocol will implement strategies to minimize the number of missing records.

Rescue therapy is anticipated to have a profound early impact on DD outcomes resulting in effectively little to no symptoms. Sensitivity analyses will be performed to support the primary analysis, using a different analytical method to handle missing data. To assess the robustness of the imputation approach using placebo as a reference for missing data due to rescue, with dilation or corticosteroids, alternative approaches will be applied: (1) A regression based approach utilizing propensity-score weighting (Funk et al. 2011); and (2) A matching based imputation approach (Abadie and Imbens 2006) where, within treatment arms, subjects who initiate rescue are matched (exactly within pre-specified covariate strata) to subjects who do not initiate rescue.

The sensitivity analyses attempt to compare treatments in the absence of rescue therapy. In the following we describe the details of simulations used to evaluate their performance under the hypothesized study conditions. Change-from-baseline outcomes and rescue therapy initiation rates are taken to be related to baseline DD and steroid-refractory status. Subjects outcomes who initiate rescue therapy are considered missing and under this simulation setting they are missing at random (MAR) since the missingness mechanism depends on their observed baseline outcomes and steroid-refractory status. The analysis methods (1) and (2) are valid under MAR conditions.

1.2 Simulation Study

Let subjects' daily diary reported dysphagia day (DD) outcomes, collected over 14-day periods, be summarized as

$$V_i = \left\{ \frac{\sum_{j=1}^{14} Y_{ij} I(Z_{ij} = 1)}{k_i} \right\} \times 14, \quad \text{if } k_i \geq 8,$$

where Y_{ij} denotes subjects ($i = 1, \dots, n$) DD outcome at day j , $I(Z_{ij} = 1)$ indicates that an outcome is recorded at day j , and $k_i = \sum_{j=1}^{14} I(Z_{ij} = 1)$ is the number of records.

The outcome V_i represents a “14-day normalized” summary of DD outcomes over a 14 day treatment period (based on at-least 8 DD records). Ideally there would be complete records for all subjects so that $k_i = 14$ for all $i = 1, \dots, n$. We refer to this as the “full data” scenario. In evaluations of stability (in terms of pooled standard deviation estimation) we find $k_i \geq 8$ to have good performance as well as limited loss of power relative to “full data”. In the sequel, we evaluate the performance of the sensitivity analyses in the presence of rescue therapy when subjects have at least 8 records and these records are randomly distributed (collected) across 1 to 14 days. We assess operating characteristics in terms of estimation and testing in comparison to the ideal “full data” scenario.

The simulations mimic the Phase 3 study design where $n = 399$ subjects will be randomized in a 2 : 1 ratio with stratification on steroid refractory status. We simulated the longitudinal nature of the Y_{ij} ’s from a binary GEE model based on the phase 2 RPC02-201 study data. Study RPC02-201 is a standard randomized, double-blind, placebo-controlled, parallel arm study that evaluates 2 doses of RPC4046 compared to Placebo for 16 weeks. 99 subjects received treatments and the study completed on 03 Jan 2017 (34 in placebo, 31 in CC-93538 180 mg SC and 34 in CC-93538 360mg SC). The primary efficacy endpoint was the change from baseline to Week 16 in the mean esophageal eosinophil count measured in the 5 most inflamed hpfs from the esophageal biopsies. Although DD was not one of the primary efficacy endpoints in the phase 2 study design, the DD outcome data was obtained by extracting patients’ responses to question 2 of Daily Symptom’s Diary (DSD) data, one of the secondary efficacy endpoints (“during any meal today, did food go down slowly or get stuck in your throat or chest?”). Table 1 provides a summary of the Phase 2 normalized outcomes (baseline and week 16) and Figure 1 displays the longitudinal DD data (Week 14 to Week 16).

We first analyzed the longitudinal DD data using the following GEE model based on an unstructured correlation specification:

$$\text{logit}(\Pr(Y_{ij} = 1)) = \beta_0 + \beta_1 j + \beta_2 V_{i0} + \beta_3 L_i + \beta_4 X_i,$$

where V_{i0} denotes subjects’ baseline V values (over 14-days prior to start of study treatment), L_i denotes steroid refractory status, and X_i denotes treatment assignment (1= experimental treatment).

We then simulated Y_{ij} ($i = 1, \dots, n, j = 1, \dots, 14$) as multivariate binary realizations with mean and correlation as estimated per the above analysis via the R package *simcormultres* (Touloumis 2016). To incorporate the Phase 3 study design assumptions, we let L_i be binary realizations where $\Pr(L_i = 1) = 0.7$ and let V_{i0} be randomly drawn from the observed (Combined) Phase 2 baseline data (Table 1). Treatment assignment X_i is randomly stratified by L_i in a 2:1 ratio.

Let $V_i(14)$ denote V_i based on “full data” DD records generated according to the above GEE model for the Y_{ij} ’s ($i = 1, \dots, n, j = 1, \dots, 14$) which induces a “full data” ANCOVA model for change from baseline:

$$V_i(14) - V_{i0} = \gamma_0 + \gamma_1 V_{i0} + \gamma_2 L_i + \gamma_3 X_i, \quad i = 1, \dots, n.$$

We conducted 10,000 simulations to mimic the study design. The number of DD records is at least 8 (randomly distributed over days 1-14) and rescue therapy is initiated at rates 15% and 5% for the control and experimental arms (respectively), however under the null (type-1 error) where both arms are identical, the experimental rescue rate is set to be the same as control. The rescue therapy initiation rates are taken to be related to baseline outcomes V_0 and steroid-refractory status via a logistic regression model (with parameters chosen to yield the overall rates per arm).

Two alternative scenarios are considered where the treatment benefit is a mean decrease in change-from-baseline, denote d , of approximately -2.0 and -1.0. We evaluate the following analyses: Ideal where complete 14-day records are available (“full data”) and NO rescue therapy is initiated; Intention-to-treat (ITT, “treatment policy”) based on at least 8 DD records where rescue therapy results in subjects’ DD outcomes ranging from 0 to one-third of their baseline DD value; Estimand analysis procedures based on a double-robust (DR) regression estimator (Funk et al. 2011) and matching analysis (denoted Matching, Abadie and Imbens 2006) wherein exact matching is conducted within categories of baseline V_0 and steroid refractory status levels. The estimand analyses treat subjects’ observed outcomes as missing if they initiate rescue therapy. Details of the analyses are described in the following section.

The key simulation studies are summarized in Tables 3-5. The estimation properties and type-1 error rates are described in Table 3, and power properties for alternative scenarios 1 (benefit of -2.0 larger decrease relative to control) and 2 (benefit of -1.0) are described in Tables 4 and 5, respectively. The methods perform well in controlling the type-1 error rates (Table 3). For a strong treatment benefit of -2.0 (Table 4) the DR and Matching analyses have power properties similar to the complete ideal scenario where complete 14-day records are available and NO rescue therapy is initiated. For a smaller treatment benefit of -1.0 (Table 5) the DR and Matching analyses have reduced power compared to the ideal scenario, but importantly, similar estimation properties. In this simulation setting, the impact of rescue therapy can significantly hamper the ability of the ITT analysis to reliably describe the treatment arms; Even under the null, while the difference is well estimated, the individual treatment means are biased where mean decreases are over-estimated due to the strong impact of rescue therapy (resulting in limited to no symptoms).

We note that there are potential challenges for the sensitivity analyses. The DR regression approach employs inverse propensity-score (PS) weighting which can be unstable when PS weights are close to zero and the Matching approach, based on exact matching within strata cells, requires for each rescue subject (in a cell) at least one non-rescue subject. For propensity score models including baseline outcomes and refractory status, we require at least 10 rescue "events per variable"; If less than 20 (but more than 10) we only include baseline V_0 ; If less than 10, we implement an intercept only model. For matching, if there are cells (formed by combinations of baseline V_0 and steroid refractory status) with insufficient counts, then we redefine categories until sufficiency (at least 1 rescue and non-rescue in each cell). These criteria were implemented in the above simulations. We note that similar findings as above were observed in additional simulations under a pathological missingness (rescue therapy) mechanism leading to "extrapolation" and matching challenges.

1.3 Analysis Details

1.3.1 DR estimator

We first describe the DR estimator of Funk et al. (2011). While the authors propose their estimator for comparing non-randomized treatments it can be readily applied to the missing data problem (See also Section 2.1 of Bang and Robins 2005). Let the observed data consist of $\{V_{i0}, L_i, \Delta_i, X_i, V_i\}$ for $i = 1, \dots, n$ where as before V_{i0} and L_i denote baseline outcome and refractory-status factors, X_i denotes treatment group, and V_i denotes the (possibly missing) outcome. Missingness is indicated by $\Delta_i = 1$ if observed and $\Delta_i = 0$ if missing. We apply the DR estimator within each treatment arm separately. First consider the estimation for the experimental arm which, for ease of notation, we consider to be the first n_1 observations (by sorting according to treatment). That is, based on data $\{V_{i0}, L_i, \Delta_i, V_i\}$ for $i = 1, \dots, n_1$, we aim to estimate the mean change-from-baseline (CBL) for the experimental arm $m_1 := E(C_1)$, where $C_1 = V - V_0$ denotes the CBL for the experimental arm. For ease of notation, let $W := (L, V_0)$ denote the set of baseline factors. We assume that C_1 is MAR where $\Pr(\Delta = 1|C_1, W) = \Pr(\Delta = 1|W) := \pi(W) > 0$. Under MAR, the mean $m_1 = E(C_1) = E[E(C_1|W)]$ can be represented in terms of the observed data as either (a) $E[E(C_1|\Delta = 1, W)]$ or (b) $E(\Delta C_1/\pi(W))$ (See Section 2.1 of Bang and Robins 2005). The DR estimator of Funk et al. (2011) is a "combination" of these representations and constructed as follows. Let $\hat{\pi}(W_i) = \hat{\pi}(W_i; \hat{\alpha})$ denote estimated probabilities based on a logistic regression model for Δ with covariate vector W (That is, $\text{logit}(\Pr(\Delta_i = 1)) = \alpha_0 + \alpha_1 V_{i0} + \alpha_2 L_i$, for $i = 1, \dots, n_1$, say) and let $\hat{m}_1(W_i) = \hat{m}_1(W_i; \hat{\gamma}) = W_i' \hat{\gamma} = \hat{\gamma}_0 + \hat{\gamma}_1 V_{i0} + \hat{\gamma}_2 L_i$ denote the "predicted response" for each subject where $\hat{\gamma}$ is the estimated ANCOVA model (linear regression) parameter vector based on the experimental subjects with observed outcomes $\Delta_i = 1$ (viz-a-viz representation (a)). The DR estimator for m_1 is defined as $\hat{m}_1 = n_1^{-1} \sum_{i=1}^{n_1} \hat{U}_{1i}$ where

$$\hat{U}_{1i} = \frac{\Delta_i C_{1i}}{\hat{\pi}(W_i)} - \frac{\{\Delta_i - \hat{\pi}(W_i)\}}{\hat{\pi}(W_i)} \hat{m}_1(W_i), \quad i = 1, \dots, n_1.$$

The variance is estimated by $\hat{\sigma}_1^2 = n_1^{-2} \sum_{i=1}^{n_1} \hat{I}_{1i}^2$ where $\hat{I}_{1i} = \hat{U}_{1i} - \hat{m}_1$ (See equation (21) of Lunceford and Davidian 2004). We note that the DR (double-robust) estimator is consistent if either the

propensity-score model $\hat{\pi}(W)$ is correct, or the outcome regression model $\hat{m}_1(W)$, for $E(C_1|\Delta = 1, W)$, is correct.

The DR estimator for the mean CBL for the control arm is calculated completely analogously as $\hat{m}_0$ with variance estimate $\hat{\sigma}_0^2$ (say). For the mean difference estimate we have $\hat{d} = \hat{m}_1 - \hat{m}_0$ with variance estimate $\hat{\sigma}_d^2 = \hat{\sigma}_1^2 + \hat{\sigma}_0^2$. Standard asymptotic 95% CI's are based on $\hat{d}$ and $\hat{\sigma}_d^2$. Let $Z = \hat{d}/\sqrt{\hat{\sigma}_d^2}$ and note that 1-sided and 2-sided p-values are $\Phi(Z)$ (as larger CBL decreases, negative values, are in favor of the experimental arm) and $2(1 - \Phi(|Z|))$, respectively, where Φ is the standard normal CDF.

1.3.2 Matching estimator

We now describe the matching estimator based on the theory of [Abadie and Imbens \(2006\)](#). The matching is carried out within each treatment arm and we first describe the approach for the experimental arm. Our goal is to compare treatments in the absence of rescue therapy, “had rescue therapy not been initiated”. Let $C_{1i}(1)$ and $C_{1i}(0)$ denote the potential outcomes for an experimental subject under no rescue and rescue therapy. For subjects with $\Delta_i = 1$ — those who did not initiate rescue therapy — we observe their potential outcome $C_{1i}(1)$ in the absence of rescue therapy; Whereas for those subjects with $\Delta_i = 0$, we do not observe their outcome $C_{1i}(1)$ and is therefore missing. We match rescue subjects to non-rescue subjects by applying 1:1 matching with replacement and allowing for ties. We use the matching algorithm of [Sekhon \(2011\)](#) as implemented in the R package *Matching*.

For matching, we use exact matching within cells formed by baseline V_0 categories and steroid-refractory status levels. For simplicity of presentation, let V_0 be categorized as $V_0 \leq 8$ and $V_0 > 8$. Then there are 4 cells: (a) $V_0 \leq 8$ and steroid-refractory; (b) $V_0 \leq 8$ and non steroid-refractory; (c) $V_0 > 8$ and steroid-refractory; and (d) $V_0 > 8$ and non steroid-refractory. Suppose that for rescue therapy subject i ($\Delta_i = 0$) they fall in cell (a) and that there are $r(i)$ non-rescue subjects (matches) in this cell. Let $\mathcal{J}(i)$ denote the indices of the $r(i)$ non-rescue subjects in cell (a). Then the outcomes for each non-rescue subject in $\mathcal{J}(i)$ is weighted by $1/r(i)$ for rescue subject i and the matching estimator imputes their missing potential outcome $C_{1i}(1)$ as $\hat{C}_{1i}(1) = r(i)^{-1} \sum_{j \in \mathcal{J}(i)} C_{1j}$. In order to carry out 1:1 matching we need at least 1 (That is, $r(i) \geq 1$) non-rescue subject in each cell that contains a rescue subject. For ease of notation let $i = 1, \dots, n_1$ denote the indices of the experimental subjects (i.e., sorting by treatment). We can then write the experimental subjects’ imputed $C_{1i}(1)$ outcome as

$$\hat{C}_{1i}(1) = \begin{cases} C_{1i}, & \text{if } \Delta_i = 1, \\ \frac{1}{r(i)} \sum_{j \in \mathcal{J}(i)} C_{1j}, & \text{if } \Delta_i = 0, \end{cases} \quad \text{for } i = 1, \dots, n_1,$$

where of course $\hat{C}_{1i}(1) = C_{1i}$ with $\Delta_i = 1$ does not require imputation as we observe their potential outcome since rescue therapy was not initiated. Notice that for each rescue subject, provided at least 1 non-rescue match exists ($r(i) \geq 1$), we have their imputed outcome and thus have a complete imputed dataset “had rescue therapy not been initiated” given by $\{\hat{C}_{1i}(1)\}_{i=1}^{n_1}$.

The matching estimator for the experimental mean CBL, in the absence of rescue therapy, is simply $\hat{m}_1 = n_1^{-1} \sum_{i=1}^{n_1} \hat{C}_{1i}(1)$. The variance estimate is more complicated and we follow the setup of [Abadie and Imbens \(2008\)](#). Recall that $\mathcal{J}(i)$ is the set of indices for the $r(i)$ non-rescue matches for rescue subject i (if subject i is a non-rescue subject then this is defined as empty): $\mathcal{J}(i) = \{j_1(i), \dots, j_{r(i)}(i)\}$. Let $K(i)$ denote the number of times subject i is used as a match: $K(i) = \sum_{l=1}^{n_1} I\{i \in \mathcal{J}(l)\}$ ($K(i)$ is zero for rescue subjects). For example, if non-rescue subject i is used as a match for 3 different rescue subjects, then $K(i) = 3$ records how often subject i contributes to the mean and variance. However, recall that each time subject i is used as a match for a rescue subject l (say) they are weighted as $\frac{1}{r(l)}$ where $r(l)$ is the number of matches for subject l . To record the weighted number of times subject i is used as a match we define

$$K_r(i) = \begin{cases} 0, & \text{if } \Delta_i = 0, \\ \sum_{l: \Delta_l = 0} I(i \in \mathcal{J}(l)) \frac{1}{r(l)}, & \text{if } \Delta_i = 1. \end{cases}$$

For example, suppose that subject i is used as a match for $K(i) = 3$ rescue subjects l_1, l_2 , and l_3 (say) with weights $r(l_1) = 1/5, r(l_2) = 1/3$, and $r(l_3) = 1/4$. The $K_r(i) = 1/5 + 1/3 + 1/4$ records their weighted contribution to the mean and variance. We can then write

$$\hat{m}_1 = n_1^{-1} \sum_{i=1}^{n_1} \hat{C}_{1i}(1) = n_1^{-1} \sum_{i=1}^{n_1} (\Delta_i + K_r(i)) C_{1i}.$$

Note that for the above subject i example, $\Delta_i = 1$ and they are used as a match for $K(i) = 3$ subjects and weighted as $K_r(i) = 1/5 + 1/3 + 1/4$. This subject therefore contributes to the mean as “himself”, $\Delta_i C_{1i} = C_{1i}$, as well as the weighted count $K_r(i) C_{1i} = (1/5 + 1/3 + 1/4) C_{1i}$. Next, define

$$K_{r^2}(i) = \begin{cases} 0, & \text{if } \Delta_i = 0, \\ \sum_{l: \Delta_l = 0} I(i \in \mathcal{J}(l)) \left\{ \frac{1}{r(l)} \right\}^2, & \text{if } \Delta_i = 1. \end{cases}$$

Then by [Abadie and Imbens \(2006\)](#) (application of their Theorem 6) the asymptotic variance of $\hat{m}_1$ can be consistently estimated by

$$\hat{\sigma}_1^2 = \frac{1}{n_1^2} \sum_{i=1}^{n_1} (\hat{C}_{1i}(1) - \hat{m}_1)^2 + \frac{1}{n_1^2} \sum_{i=1}^{n_1} (K_r^2(i) + \Delta_i + 2\Delta_i K_r(i)) \hat{\zeta}^2(\Delta_i),$$

where, for brevity, we note that we use the estimator “se.standard” from the R package *Matching* ([Sekhon 2011](#)) for $\hat{\zeta}^2(\Delta_i)$ which is the conditional variance of C_{1i} (conditional on Δ_i and the matching strata formed by baseline V_{0i} and steroid-refractory status).

Here we use the notation $\hat{m}_1$ and $\hat{\sigma}_1^2$ generically for both the DR and matching estimator for the experimental mean CBL, $m_1 = E(C_1)$, and its variance (similarly for control and their difference).

The matching estimator for the mean CBL for the control arm is calculated completely analogously as $\hat{m}_0$ with variance estimate $\hat{\sigma}_0^2$ (say). For the mean difference estimate we have $\hat{d} = \hat{m}_1 - \hat{m}_0$ with variance estimate $\hat{\sigma}_d^2 = \hat{\sigma}_1^2 + \hat{\sigma}_0^2$. As for the DR estimator, 1-sided and 2-sided p-values are $\Phi(Z)$ and $2(1 - \Phi(|Z|))$ where $Z = \hat{d} / \sqrt{\hat{\sigma}_d^2}$.

References

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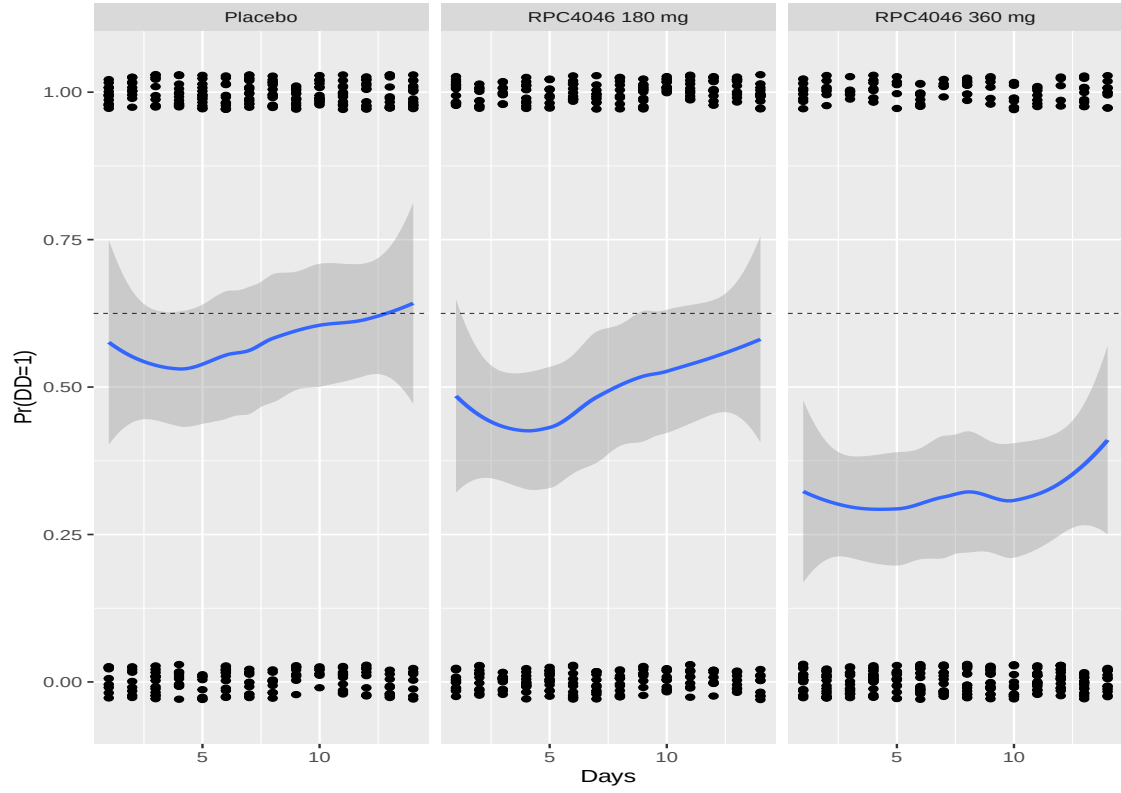


Fig. 1: Phase 2 Week 14 to Week 16 DD records by treatment group. The solid curves denote loess fits along with pointwise 95% confidence intervals (dark grey shaded region).

Table 1: Phase-2 baseline (dd14.bl) and Week 16 (dd14) dysplasia days normalized to 14-days

	N	Placebo			RPC4046 180 mg			RPC4046 360 mg			Combined		
		<i>N</i> = 34			<i>N</i> = 31			<i>N</i> = 34			<i>N</i> = 99		
dd14.bl	99	8.8	10.3	14.0	8.8	12.4	14.0	9.0	11.8	14.0	8.8	12.0	14.0
dd14	75	2.52	7.88	14.00	0.29	6.89	13.75	0.00	1.27	9.33	0.00	5.25	13.50

a b c represent the lower quartile a , the median b , and the upper quartile c for continuous variables. N is the number of non-missing values.

Table 2: Null model (Type-1 error) based on 10,000 simulations: Underlying (true) experimental and control change-from-baseline means are identical (approximately -2.1) and rescue therapy rates are both 0.15. The true difference is $d_0=0$.

m1	m0	CP1	CP0	d	SE	ESE	CP	R1	R2
Ideal ITT in absence of rescue									
-2.082	-2.087	0.956	0.962	0.005	0.298	0.302	0.953	0.022	0.047
ITT in presence of rescue									
-3.284	-3.287	0.001	0.075	0.002	0.431	0.431	0.951	0.024	0.049
DR estimand analysis									
-2.081	-2.090	0.951	0.961	0.010	0.335	0.364	0.967	0.015	0.034
Matching estimand analysis									
-2.080	-2.094	0.971	0.977	0.014	0.345	0.414	0.982	0.008	0.019

m1 (m0), avg. of estimated experimental (control) means; d, avg. of estimated differences; SE, standard error (SE) of estimated differences; ESE, avg. of estimated SEs; CP, confidence-interval coverage probabilities for means (CP1 (CP0) for experimental (control), CP for difference); R1 and R2 are 1-sided (superiority) and 2-sided rejection rates.

Table 3: Alternative 1 model (Power) based on 10,000 simulations: Underlying (true) experimental and control change-from-baseline means are approximately -4.1 and -2.1, respectively. The rescue therapy rates are 0.15 for control and 0.05 for experimental. The true difference is $d_0=-2.0$.

m1	m0	CP1	CP0	d	SE	ESE	CP	R1	R2
Ideal ITT in absence of rescue									
-4.119	-2.087	0.957	0.962	-2.034	0.305	0.315	0.958	1.000	1.000
ITT in presence of rescue									
-4.424	-3.287	0.723	0.075	-1.138	0.409	0.386	0.366	0.822	0.822
DR estimand analysis									
-4.118	-2.090	0.960	0.961	-2.028	0.336	0.366	0.965	1.000	1.000
Matching estimand analysis									
-4.127	-2.094	0.983	0.977	-2.033	0.345	0.426	0.982	1.000	1.000

m1 (m0), avg. of estimated experimental (control) means; d, avg. of estimated differences; SE, standard error (SE) of estimated differences; ESE, avg. of estimated SEs; CP, confidence-interval coverage probabilities for means (CP1 (CP0) for experimental (control), CP for difference); R1 and R2 are 1-sided (superiority) and 2-sided rejection rates.

Table 4: Alternative 2 model (Power) based on 10,000 simulations: Underlying (true) experimental and control change-from-baseline means are approximately -3.1 and -2.1, respectively. The rescue therapy rates are 0.15 for control and 0.05 for experimental. The true difference is $d_0=-1.0$.

m1	m0	CP1	CP0	d	SE	ESE	CP	R1	R2
Ideal ITT in absence of rescue									
-3.133	-2.087	0.958	0.962	-1.047	0.302	0.310	0.954	0.924	0.924
ITT in presence of rescue									
-3.488	-3.287	0.649	0.075	-0.200	0.411	0.388	0.415	0.088	0.098
DR estimand analysis									
-3.132	-2.090	0.954	0.961	-1.042	0.333	0.364	0.966	0.831	0.831
Matching estimand analysis									
-3.140	-2.094	0.982	0.977	-1.047	0.343	0.421	0.984	0.737	0.737

m1 (m0), avg. of estimated experimental (control) means; d, avg. of estimated differences; SE, standard error (SE) of estimated differences; ESE, avg. of estimated SEs; CP, confidence-interval coverage probabilities for means (CP1 (CP0) for experimental (control), CP for difference); R1 and R2 are 1-sided (superiority) and 2-sided rejection rates.